TOPIC 6 - MIGRATING INTO THE CLOUD

Concepts, Challenges, Integration
& Real-world Considerations

CLOUD MIGRATION

- Cloud migration is the process of moving digital business operations into the cloud.
- Involves moving data, applications, and IT processes from on-premises or legacy infrastructure to a cloud-based environment.
- Can be done partially (hybrid) or fully (public/private cloud).

REASONS TO MIGRATE TO THE CLOUD

Scalability: Easily increase or decrease resources.

Cost Efficiency: Pay-as-you-go pricing reduces capital expenditure.

Performance: Improved speed, performance, and uptime.

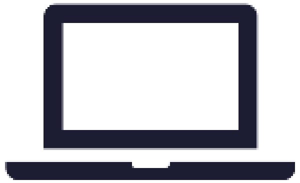
Disaster Recovery: Built-in redundancy and failover options.

Remote Access: Access from anywhere using internet.

TYPES OF CLOUD SERVICES

IaaS (Infrastructure as a Service): Provides virtualized computing resources via the internet (e.g., AWS EC2, Azure VM).

PaaS (Platform as a Service): Offers hardware and software tools for application development (e.g., Google App Engine).



SaaS (Software as a Service): Delivers software applications over the internet (e.g., Google Workspace, Dropbox).

POPULAR CLOUD PROVIDERS

- **Amazon Web Services (AWS):** Leader in global cloud market.
- **Microsoft Azure:** Strong in hybrid cloud integration.
- **Google Cloud Platform (GCP):** Known for data analytics and AI tools.
- **IBM Cloud, Oracle Cloud:** Used in enterprise and legacy migration contexts.

KEY CHALLENGES IN CLOUD MIGRATION

DOWNTIME: RISK OF APPLICATION UNAVAILABILITY DURING MIGRATION.

DATA LOSS: IMPROPER HANDLING CAN LEAD TO PERMANENT DATA LOSS.

SECURITY: DATA IN TRANSIT AND AT REST MUST BE PROTECTED.

COMPLIANCE: MUST MEET REGULATORY REQUIREMENTS LIKE GDPR, HIPAA.

COMPATIBILITY: LEGACY APPS MAY NOT RUN SMOOTHLY IN CLOUD.

PLANNING

Business Goals: Define clear objectives (e.g., cost reduction, scalability).

Assessment: Evaluate existing infrastructure and dependencies.

Cost-Benefit Analysis: Weigh potential savings against migration costs.

Choose Cloud Model: Public, private, or hybrid?

THE 6 R’S OF CLOUD MIGRATION

Rehost	Lift and shift.
Refactor	Minor code changes to fit cloud environment.
Revise	Major changes to existing code.
Rebuild	Redesign from scratch.
Replace	Substitute with SaaS solutions.
Retire	Eliminate outdated components.

REHOST (LIFT AND SHIFT)

- Move applications without modifying code.
- Typically faster and less expensive.
- Suitable for legacy systems that are hard to refactor.

- **Example:** Moving a virtual machine from a local data center to AWS EC2.

REFACTOR (REPLATFORM)

- Make minimal changes to adapt the app to the cloud.
- Improve performance or scalability.
- Reuse existing code and frameworks.

Example: Moving an app to a managed database or using container services.

REVISE (RE-ARCHITECT)

- Modify or extend existing code base.
- Required when existing code is not optimized for cloud.
- Prepares for better long-term cloud-native benefits.

Example: Splitting a monolithic app into microservices.

REBUILD (REDESIGN FROM SCRATCH)

- Complete rewrite of the application.
- Highest cost and time investment.
- Best option for outdated or inefficient legacy systems.

Example: Rebuilding an HR system using cloud-native frameworks.

REPLACE (DROP AND SHOP)

- Replace your in-house app with SaaS.
- Avoids development cost and effort.
- Fast deployment and maintenance.

Example: Switching to Salesforce instead of running your own CRM.

RETIRE (DECOMMISSIONING)

- Identify and eliminate applications no longer needed.
- Frees up resources.
- Reduces complexity and cost.

Example: Retiring an old reporting tool replaced by BI dashboards in cloud.

STEPS TO CLOUD MIGRATION

CREATE A MIGRATION STRATEGY

EVALUATE APPLICATIONS AND DATA

CHOOSE THE RIGHT CLOUD PROVIDER

PREPARE THE ENVIRONMENT

MIGRATE AND TEST

MONITOR AND OPTIMIZE

STEP 1 - CREATE A MIGRATION STRATEGY

Define the business objectives for moving to the cloud.

Determine scope, timeline, and budget.

Identify stakeholders and assign responsibilities.

STEP 2 - EVALUATE APPLICATIONS AND DATA

Audit existing infrastructure.

Classify applications based on criticality and complexity.

Identify dependencies and compliance requirements.

STEP 3 - CHOOSE THE RIGHT CLOUD PROVIDER

Compare pricing models, SLAs, and services offered.

Consider support, security features, and compliance certifications.

Evaluate existing provider integrations and ecosystem compatibility.

STEP 4 - PREPARE THE ENVIRONMENT

Set up

- Set up accounts, access policies, and permissions.

Design

- Design network architecture and storage.

Implement

- Implement identity and access management (IAM).

STEP 5 - MIGRATE AND TEST

Begin with low-risk applications as a pilot.

Execute the migration using tools and automation.

Validate the system functionality through rigorous testing.

STEP 6 - MONITOR AND OPTIMIZE

Monitor usage, performance, and security.

Tune systems for cost and performance efficiency.

Implement alerts and analytics dashboards.

MIGRATION TOOLS & PLATFORMS

- **AWS Migration Hub**
- **Azure Migrate**
- **Google Migrate for Compute Engine**
- **CloudEndure, Velostrata**

LIMITATIONS THAT DETER ADOPTION

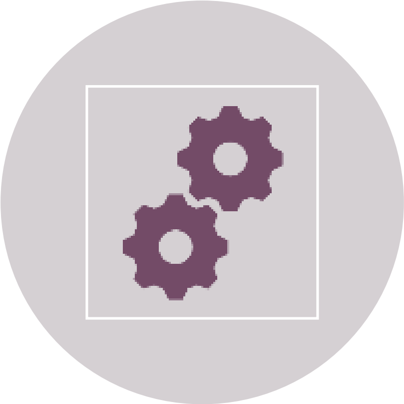
Poor Connectivity: Limits real-time access.

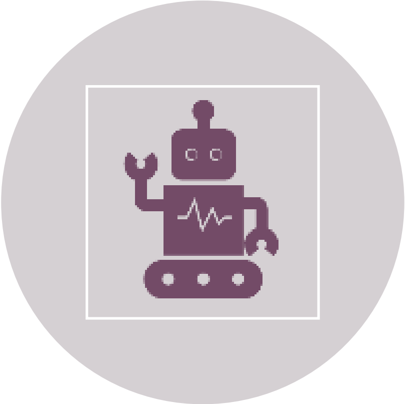
Legacy Dependencies: Older apps may resist change.

Lack of Cloud Talent: Skilled professionals are in demand.

Concerns over Vendor Lock-in

CLOUD INTEGRATION TYPES





DATA INTEGRATION:
SYNCHRONIZE DATA ACROSS SYSTEMS.

APPLICATION INTEGRATION:
CONNECT DISPARATE APPS FOR WORKFLOW.

PROCESS INTEGRATION:
END-TO-END BUSINESS PROCESS AUTOMATION.

DATA INTEGRATION

- Combines data from different sources into a single, unified view.
- Supports real-time or batch synchronization between databases and storage.
- Ensures consistency, quality, and availability of business-critical data.

Example: Syncing product inventory data between a cloud-based e-commerce platform and an on-premises ERP system.

APPLICATION INTEGRATION

- Connects different applications to work together.
- Facilitates communication and data exchange across platforms.
- Enables automation of workflows that span multiple systems.

Example: Integrating Salesforce (CRM) with Mailchimp (Email Marketing) to auto-send customer emails after a sales update.

PROCESS INTEGRATION

- Aligns business processes across applications and departments.
- Streamlines multi-step workflows that depend on input/output from different systems.
- Often managed through orchestration or Business Process Management (BPM) tools.

Example: An order placed on an e-commerce platform triggers inventory check, invoicing, and shipment tracking—spanning several apps.

REQUIREMENTS FOR INTEGRATION

- Secure APIs
- Real-time and Batch Processing Support
- Data Transformation
- Monitoring and Logging

ATTRIBUTES OF INTEGRATION PLATFORMS

Scalability:

- Must handle growing data.

Interoperability:

- Work with multiple environments.

Security:

- Data encryption and compliance support.

Ease of Use:

- Low-code/drag-drop preferred.

WHAT IS JITTERBIT INTEGRATION SERVER?

Low-code iPaaS (Integration Platform as a Service)

Supports connecting cloud and on-premise systems.

Visual data mapping and transformation.

Real-time and scheduled integrations.

RISKS IN CLOUD MIGRATION

- Vendor Lock-in
- Budget Overruns
- Service Disruptions
- Loss of Governance
- Incompatibility

MITIGATION STRATEGIES

- Use Open Standards/APIs
- Document Everything
- Start with Pilot Projects
- Regular Testing and Backup

POST-MIGRATION CONSIDERATIONS

MONITORING AND LOGGING

PERFORMANCE OPTIMIZATION

SECURITY POLICY UPDATES

USER TRAINING

PERIODIC AUDITS

CLOUD MESSAGING QUEUING

Message Queuing: Temporarily holds messages for processing.

Decouples Components: Producer and consumer run independently.

Improves Reliability: Handles spikes without overloading systems.

EXAMPLES OF MESSAGING QUEUING TOOLS

Amazon SQS: Reliable queue service for distributed apps.

Apache Kafka: High-throughput distributed messaging.

RabbitMQ: Lightweight and flexible broker.

Azure Queue Storage

SUMMARY OF KEY POINTS

CLOUD MIGRATION IS COMPLEX BUT REWARDING

CHOOSE THE RIGHT MODEL AND TOOLS

INTEGRATE THOUGHTFULLY

BE AWARE OF RISKS AND MITIGATE THEM

USE MESSAGING QUEUES FOR BETTER ARCHITECTURE